Inflation Dynamics - Computational Notebook

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Abstract

This book presents computation details of estimating inflation dynamics.

This notebook presents computation details of estimating inflation dynamics.

So far we have covered estimation of

1. GARCH
2. EGARCH
3. GJR-GARCH
4. SPARCH
5. Regime-switching GARCH
6. BEGE GARCH

State variable definitions:

1. π_t: inflation.
2. $\mathcal{F}_{t-1}$: filtration at time $t-1$.
3. SPF_{t-1}: professional forecast on π_t inflation measured by $\mathcal{F}_{t-1}$.
4. $\hat{\pi}_t$: expected inflation $E[\pi_t \mid \mathcal{F}_{t-1}]$.
5. u_t: inflation innovation (residual), defined as $u_t := \pi_t - \hat{\pi}_t$.
6. u_t^+ : postive residual, defined as $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$
7. u_t^- : negative residual, defined as $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$

Computation Notebook

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1. Inflation Summary Statistics

I start with simple statistical summary by plotting inflation and professional forecast over time.

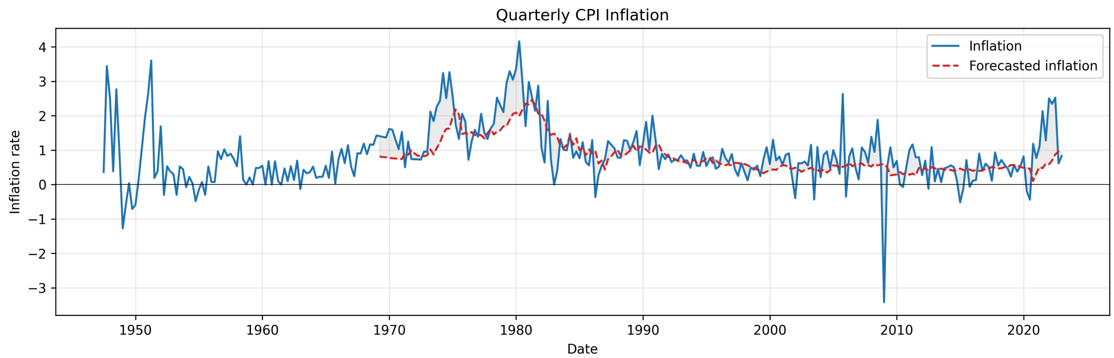


Figure 1: Quarterly inflation π_t and profesional forcast SPF_{t-1}.

Notice that inflation was recorded from 1947 but profesional forecast started from 1969.

	Inflation	Forecasted inflation
Date Start	1947Q2	1969Q1
Date End	2022Q4	2022Q4
#Observations	303	216

Table 1: Inflation statistics

1.a. Effective Sample

To make log-likelihood values comparable across specifications, I use an effective sample for all mean models and all volatility models.

In ARX models, lagged regressors (for example, SPF_{t-1}) are not observed at the very beginning of the raw sample. Therefore, I start the estimation sample at the first date where both current forecast inflation (SPF_t), and one-lag forecast inflation (SPF_{t-1}) are available. For the quarterly data, forecast inflation (SPF_t) is available from **1969Q1** to **2022Q4** (216 observations), requiring one lag (SPF_{t-1}) shifts the usable start to **1969Q2**. So the trimmed estimation window is **1969Q2–2022Q4**, with **215 observations**. This same trimmed sample is then used for all mean specifications, so the residual series has the same length across models.

For GARCH-type recursion, the first few conditional variances are unobserved (for example, σ_0^2, σ_{-1}^2, u_0^2, u_{-1}^2). I initialize these pre-sample variances using the model-implied unconditional variance, for example in GARCH model:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \sum_i \alpha_i - \sum_j \beta_j}, \tag{1}$$

and set

$$u_0^2 = u_{-1}^2 = \cdots = \sigma_0^2 = \sigma_{-1}^2 = \cdots = \bar{\sigma}^2. \tag{2}$$

1.b. Effective Sample Summary Statistics

I report the summary statistics for trimmed effect sample. Skewness and Kurtosis are adjusted by sample size.

	Inflation (π)	SPF	$\pi - SPF$
Date Start	1969Q2	1969Q2	1969Q2
Date End	2022Q4	2022Q4	2022Q4
#Observations	215	215	215
Mean	0.9918	0.8320	0.1598
Median	0.7937	0.6296	0.1101
Std	0.8580	0.4969	0.6642
Min	−3.4170	0.1048	−4.0117
Max	4.1612	2.4566	2.1729
P5	−0.0681	0.3482	−0.7745
P25	0.5328	0.4748	−0.1669
P75	1.2913	1.0150	0.4617
P95	2.5883	2.0030	1.2581
Skewness	0.3550	1.3633	−0.7131
Kurtosis	3.8995	1.2053	7.3006
AC(1)	0.5981	0.9717	0.2764
AC(2)	0.5353	0.9473	0.1322
AC(4)	0.4708	0.8929	0.0881
AC(12)	0.2918	0.6949	0.0011

Table 2: Inflation statistics

2. Mean Process Specification

Four types of mean processes are of interest:

- **Constant**: $\pi_{t+1} = SPF_t + \mu_{t+1}$
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,2)**[1]: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + \mu_{t+1}$

where μ_{t+1} is the residual term.

2.a. OLS Results

Estimation results of four mean models with HAC standard errors in parentheses below each estimate. (The log-likelihood is computed under the Gaussian assumption.)

Constant:

$$\hat{\pi}_{t+1} = SPF_t \tag{3}$$

No estimated coefficients. Residuals $\mu_{t+1} = \pi_{t+1} - SPF_t$.

- Log-likelihood: −**222.664**
- AIC: **447.329**, BIC: **450.700**
- Residual skewness: −**0.713**, excess kurtosis: **7.301**

ARX(1,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0824} \underset{(0.112)}{+0.3005\,\pi_t} \underset{(0.167)}{+0.7337\,SPF_t} \tag{4}$$

- Log-likelihood: −**207.381**
- AIC: **420.762**, BIC: **430.874**
- Residual skewness: −**1.047**, excess kurtosis: **8.210**

ARX(2,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0897} \underset{(0.098)}{+0.2892\,\pi_t} \underset{(0.126)}{+0.0834\,\pi_{t-1}} \underset{(0.251)}{+0.6385\,SPF_t} \tag{5}$$

- Log-likelihood: −**206.810**
- AIC: **421.619**, BIC: **435.102**
- Residual skewness: −**1.191**, excess kurtosis: **9.050**

ARX(2,2):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0856} \underset{(0.106)}{+0.2992\,\pi_t} \underset{(0.125)}{+0.0914\,\pi_{t-1}} \underset{(0.433)}{+0.4084\,SPF_t} \underset{(0.297)}{+0.2136\,SPF_{t-1}} \tag{6}$$

- Log-likelihood: −**206.660**
- AIC: **423.319**, BIC: **440.172**
- Residual skewness: −**1.212**, excess kurtosis: **9.129**

[1] In the GARCH family estimation exercise, we found that ARX(2,2) is always dominated by other three specifications. Thus, we stop estimating ARX(2,2) in BEGE estimation to saving computational resources.

3. GARCH Family

This section presents three GARCH-type volatility models — GARCH, GJR-GARCH, and EGARCH — each combined with one of three conditional distributions for residuals: normal, Student's t, and a finite mixture of normals.

Let $\{u_t\}_{t=1}^{T}$ denote the residual process, and let $\mathcal{F}_{t-1}$ denote the information set available at time $t-1$. We assume

$$u_t = \sigma_t \, z_t, \qquad z_t \mid \mathcal{F}_{t-1} \overset{\text{i. i. d.}}{\sim} D(0,1), \tag{7}$$

where $\sigma_t^2 = \operatorname{Var}(u_t \mid \mathcal{F}_{t-1})$ is the conditional variance and $D(0,1)$ is a standardized distribution (normal, Student's t, or mixture of normals) with zero mean and unit variance. Throughout, $z_t = u_t / \sigma_t$ denotes the standardized residual and θ denotes the full parameter vector (mean process, volatility process parameters together with any distributional parameters).

3.a. *Volatility Recursion Specifications*

GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \, u_{t-i}^2 + \sum_{k=1}^{q} \beta_k \, \sigma_{t-k}^2, \tag{8}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\beta_k \geq 0$, and $\sum_i \alpha_i + \sum_k \beta_k < 1$ for stationarity.

GJR-GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(u_{t-i}^{+}\right)^2 + \sum_{j=1}^{p} \gamma_j \left(u_{t-j}^{-}\right)^2 + \sum_{k=1}^{q} \beta_k \, \sigma_{t-k}^2, \tag{9}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\gamma_j \geq 0$, and $\beta_k \geq 0$. The asymmetry coefficient γ_j captures the leverage effect: when $\alpha_j > \gamma_j$, positive shocks raise future volatility more than negative shocks of equal magnitude.

Under the assumption that z_t is symmetric

$$\mathbb{E}\left[\left(u_{t-i}^{+}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \mathbb{E}\left[\left(u_{t-i}^{-}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \frac{1}{2} \sigma_{t-i}^2 \tag{10}$$

, the stationarity condition is

$$\frac{1}{2} \sum_{i=1}^{p} \alpha_i + \frac{1}{2} \sum_{j=1}^{p} \gamma_j + \sum_{k=1}^{q} \beta_k < 1. \tag{11}$$

EGARCH(p, q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i (\mid z_{t-i} \mid - \mathbb{E} \mid z_{t-i} \mid) + \sum_{j=1}^{p} \gamma_j \, z_{t-j} + \sum_{k=1}^{q} \beta_k \ln \sigma_{t-k}^2, \tag{12}$$

where $z_t = u_t / \sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks), while γ_j captures the sign / leverage effect. Modeling $\ln \sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $\left(\alpha_i, \gamma_j, \beta_k\right)$, and the recursion is covariance-stationary in $\ln \sigma_t^2$ whenever

$$\sum_{k=1}^{q} \beta_k < 1. \tag{13}$$

The constant $\mathbb{E} \mid z_t \mid$ depends on the assumed distribution of the standardized innovation:

- **Standard normal**, $z_t \sim \mathcal{N}(0, 1)$:

$$\mathbb{E} \mid z_t \mid = \sqrt{\frac{2}{\pi}}. \tag{14}$$

- **Standardized Student's t** with $\nu > 2$ degrees of freedom (rescaled to unit variance):

$$\mathbb{E} \mid z_t \mid = \frac{2\sqrt{\nu - 2}\,\Gamma\left(\frac{\nu+1}{2}\right)}{(\nu - 1)\sqrt{\pi}\,\Gamma\left(\frac{\nu}{2}\right)}. \tag{15}$$

As $\nu \to \infty$ this collapses to the normal value $\sqrt{2/\pi}$; for finite ν it is strictly larger, reflecting the heavier tails.

- **Mixture of two normals** with parameters $\left(p_1, \mu_1, \sigma_1^2\right)$ and induced $\left(p_2, \mu_2, \sigma_2^2\right)$ (as defined in the likelihood section): using $\mathbb{E} \mid X \mid = \mu\,[2\Phi(\mu/s) - 1] + 2s\,\phi(\mu/s)$ for $X \sim \mathcal{N}\left(\mu, s^2\right)$ (a folded-normal mean), where Φ and ϕ are the standard normal CDF and PDF,

$$\mathbb{E} \mid z_t \mid = \sum_{m=1}^{2} p_m\left[\mu_m(2\Phi(\mu_m/\sigma_m) - 1) + 2\sigma_m\,\phi(\mu_m/\sigma_m)\right]. \tag{16}$$

EGARCH(p, q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i\left(\mid z_{t-i} \mid - \sqrt{\frac{2}{\pi}}\right) + \sum_{j=1}^{p} \gamma_j\, z_{t-j} + \sum_{k=1}^{q} \beta_k \ln \sigma_{t-k}^2, \tag{17}$$

where $z_t = u_t/\sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks) while γ_j captures the sign / leverage effect. Modeling $\ln \sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $\left(\alpha_i, \gamma_j, \beta_k\right)$, and the recursion is covariance-stationary in $\ln \sigma_t^2$ whenever

$$\sum_{k=1}^{q} \beta_k < 1. \tag{18}$$

Remark on the centering constant. The constant $\sqrt{2/\pi}$ equals $\mathbb{E} \mid z_t \mid$ when $z_t \sim \mathcal{N}(0, 1)$, so under a Gaussian innovation the term $\mid z_{t-i} \mid - \sqrt{2/\pi}$ has zero conditional mean and ω admits the clean interpretation

$$\mathbb{E}[\ln \sigma_t^2] = \frac{\omega}{1 - \sum_{k=1}^{q} \beta_k}. \tag{19}$$

Following the convention of the `arch` Python package, we retain $\sqrt{2/\pi}$ as a fixed constant in the recursion even when the innovation distribution is Student's t or a mixture of normals. Under those distributions, $\mathbb{E} \mid z_t \mid \neq \sqrt{2/\pi}$, so the centered term $\mid z_{t-i} \mid - \sqrt{2/\pi}$ no longer has zero mean and ω loses the unconditional-mean interpretation above. This choice is observationally innocuous: replacing $\sqrt{2/\pi}$ with the distribution-specific constant $c_D = \mathbb{E}_D \mid z_t \mid$ yields an equivalent model whose fitted $\hat{\alpha}_i, \hat{\gamma}_j, \hat{\beta}_k$ are unchanged, with only the intercept shifted by

$$\omega = \omega^* + \left(\sqrt{\frac{2}{\pi}} - c_D\right) \sum_{i=1}^{p} \alpha_i. \tag{20}$$

The likelihood, the fitted σ_t^2 path, and all forecasts are identical under the two parameterizations.

3.b. Log-Likelihood Specifications

For each candidate distribution we give (i) the standardized density of z_t, (ii) the conditional density of $u_t \mid \mathcal{F}_{t-1}$ obtained via the change of variables $u_t = \sigma_t z_t$ (which contributes a Jacobian factor $1/\sigma_t$), (iii) the per-observation log-likelihood $\ell_t(\theta)$, and (iv) the sample log-likelihood

$$\ln L(\theta) = \sum_{t=1}^{T} \ell_t(\theta). \tag{21}$$

3.b.i. *Normal distribution:*

The standardized innovation satisfies $z_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, 1)$ with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\}. \tag{22}$$

By the change of variables $u_t = \sigma_t z_t$, the conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} \phi\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sqrt{2\pi\, \sigma_t^2}} \exp\left\{-\frac{u_t^2}{2\, \sigma_t^2}\right\}. \tag{23}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\frac{1}{2}\left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right), \tag{24}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\frac{1}{2} \sum_{t=1}^{T} \left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right). \tag{25}$$

3.b.ii. *Student's t distribution:*

Let $\nu > 2$ denote the degrees-of-freedom parameter. The standardized Student's t density (rescaled to unit variance) is

$$f_\nu(z) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)}} \left(1 + \frac{z^2}{\nu - 2}\right)^{-\frac{\nu+1}{2}}, \tag{26}$$

where $\Gamma(\cdot)$ is the gamma function. The restriction $\nu > 2$ ensures finite variance. The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} f_\nu\left(\frac{u_t}{\sigma_t}\right) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)\, \sigma_t^2}} \left(1 + \frac{u_t^2}{(\nu-2)\, \sigma_t^2}\right)^{-\frac{\nu+1}{2}}. \tag{27}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = \ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\sigma_t^2\ \right) - \frac{\nu+1}{2}\ln\left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right), \tag{28}$$

and the sample log-likelihood is

$$\ln L(\theta) = T\left[\ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\ \right)\right] - \frac{1}{2}\sum_{t=1}^{T}\ln\sigma_t^2 - \frac{\nu+1}{2}\sum_{t=1}^{T}\ln\left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right). \tag{29}$$

3.b.iii. *Mixture of two normals:*

The standardized innovation follows a two-component Gaussian mixture with parameters (p_1, μ_1, σ_1^2):

$$f_{\mathrm{mix}}(z) = p_1\,\phi(z;\mu_1,\sigma_1^2) + p_2\,\phi(z;\mu_2,\sigma_2^2), \qquad \phi(z;\mu,s^2) = \frac{1}{\sqrt{2\pi s^2}}\exp\left\{-\frac{(z-\mu)^2}{2s^2}\right\}. \tag{30}$$

To enforce $\mathbb{E}[z_t] = 0$ and $\mathrm{Var}(z_t) = 1$, the remaining parameters are determined by

$$p_2 = 1 - p_1, \qquad \mu_2 = -\frac{p_1\mu_1}{p_2}, \qquad \sigma_2^2 = \frac{1 - p_1(\sigma_1^2 + \mu_1^2) - p_2\mu_2^2}{p_2}. \tag{31}$$

This leaves three free parameters: (p_1, μ_1, σ_1^2).

Hierarchical representation. The mixture admits the latent-variable form $S \sim \mathrm{Bernoulli}(p_1)$, $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, with $z = X_1$ if $S = 1$ and $z = X_2$ otherwise. As a consequence, moments and the CDF decompose as

$$\mathbb{E}[z^n] = p_1\,\mathbb{E}[X_1^n] + p_2\,\mathbb{E}[X_2^n], \qquad F_{\mathrm{mix}}(a) = p_1\,\Phi\left(\frac{a-\mu_1}{\sigma_1}\right) + p_2\,\Phi\left(\frac{a-\mu_2}{\sigma_2}\right), \tag{32}$$

where Φ is the standard normal CDF.

The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1};\theta) = \frac{1}{\sigma_t} f_{\mathrm{mix}}\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sigma_t}\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right], \qquad z_t = \frac{u_t}{\sigma_t}. \tag{33}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\ln\sigma_t + \ln\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right], \tag{34}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\sum_{t=1}^{T}\ln\sigma_t + \sum_{t=1}^{T}\ln\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right]. \tag{35}$$

3.b.iv. *Gaussian Mixture distribution:*

The mixture of 2 normal distribution given parameters p_1,μ_1,σ_1^2 is

$$f(z_t) = p_1\frac{1}{\sqrt{2\pi\sigma_1^2}}exp\left\{-\frac{\left(z_t-\mu_1\right)^2}{2\sigma_1^2}\right\} + p_2\frac{1}{\sqrt{2\pi\sigma_2^2}}exp\left\{-\frac{\left(z_t-\mu_2\right)^2}{2\sigma_2^2}\right\} \tag{36}$$

where $p_2 = 1 - p_1$,$\mu_2 = \frac{-p_1\mu_1}{p_2}$, and $\sigma_2^2 = \frac{1-p_2\mu_2^2-p_1(\sigma_1^2+\mu_1^2)}{p_2}$

N th order moment can be calculated as:

$$E[z^n] = \int z^n \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z-\mu_1)^2}{2\sigma_1^2} \right\} + (1-p_1) \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z-\mu_2)^2}{2\sigma_2^2} \right\} \right] dz \quad (37)$$

$$E[z^n] = p_1 E[x_1^n \mid x_1 \sim N(\mu_1, \sigma_1^2)] + p_2 E[x_2^n \mid x_2 \sim N(\mu_2, \sigma_2^2)] \quad (38)$$

Similary, CDF can be calculated analytically by

$$\Phi(a) = \int_{-\inf}^{a} z \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z-\mu_1)^2}{2\sigma_1^2} \right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z-\mu_2)^2}{2\sigma_2^2} \right\} \right] dz \quad (39)$$

$$\Phi(a) = p_1 \Phi_{x_1}(a) + p_2 \Phi_{x_2}(a) \quad (40)$$

We can think mixture of two normal distributions generated by three random variable Z,X_1,X_2. Z follows Bernoulli(P_1), $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$. In order to generate a random sample point, we can first use Z to decide which normal distributions to use, and then randomly pick a point in that normal distribution.

Log likelihood function is

$$L(u_t) = \sum_{t=1}^{T} \ln \frac{1}{\sigma_t} \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z_t-\mu_1)^2}{2\sigma_1^2} \right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z_t-\mu_2)^2}{2\sigma_2^2} \right\} \right] \quad (41)$$

, where $z_t = \frac{u_t}{\sigma_t}$

3.c. Estimation Summary

3.d. Best Estimation Illustrations

Table 2: Model selection by AIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	359.4710	(1,1)	(1,1)	368.2429
t	(1,1)	(2,1)	343.1431	(2,1)	(1,1)	344.6254
Skew *t*	(1,1)	(2,1)	345.1312	(2,1)	(1,1)	346.5791
GED	(1,1)	(2,1)	345.1925	(1,1)	(1,1)	347.6934
Mix of Normal	(1,1)	(2,1)	345.8222	(2,1)	(2,1)	343.5023

Table 3: Model selection by BIC: GJR vs EGARCH

Distribution	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	381.4280	FC	(1,1)	383.4754
t	FC	(1,1)	367.8350	FC	(1,1)	367.1732
Skew *t*	FC	(1,1)	370.4242	FC	(1,1)	369.6649
GED	FC	(1,1)	371.8489	FC	(1,1)	370.9644
Mix of Normal	FC	(1,1)	373.7106	FC	(1,1)	371.0689

4. Model Selection Report

Table 2: Model selection by AIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH AIC	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	387.6789	(1,1)	(2,1)	372.8425	(1,1)	(2,1)	374.2995
t	(2,1)	(2,1)	362.6718	(1,1)	(2,1)	357.3153	(1,1)	(2,1)	356.7451
Mix of Normal	(2,1)	(1,2)	365.6060	(2,1)	(2,1)	359.0690	(1,1)	(2,1)	356.4744

Table 3: Model selection by BIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH BIC	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	401.4156	FC	(2,1)	396.6581	FC	(2,1)	398.1066
t	FC	(2,1)	387.7770	FC	(1,1)	384.2113	FC	(1,1)	383.8115
Mix of Normal	(1,1)	(2,1)	401.6779	FC	(2,1)	391.9670	FC	(1,1)	385.2567

4.a. *Best Models by Criterion and Volatility Family*

For each criterion and each volatility family, the selected model is the best across mean-process choices, orders, and distributions using stable & successful fits.

4.a.i. *AIC:*

GARCH:

- Model ID: `M079`
- Distribution: `$t$`
- Mean process: `(2,1)` (ARX_2_1)
- Volatility process: `GARCH(2,1)`

$$\hat{\pi}_{t+1} = 0.0754 + 0.2410\,\pi_t + 0.1539\,\pi_{t-1} + 0.5650\,SPF_t + \mu_{t+1} \tag{42}$$

Robust SE: `Const` (0.0804), `Inflation_lag_1` (0.1565), `Inflation_lag_2` (0.2555), `SPF` (0.4729).

$$\hat{\sigma}_t^2 = 0.1369 + 0.4194\,u_{t-1}^2 + 0.3468\,u_{t-2}^2 + 0.0588\,\sigma_{t-1}^2 \tag{43}$$

- Number of observations: **215**
- Log-likelihood: **−172.335915**
- AIC: **362.671831**, BIC: **393.007573**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.075394	0.080383	0.937930	0.348280
Inflation_lag_1	0.241000	0.156521	1.539733	0.123625

Parameter	Coef	Std Err	t-value	p-value
Inflation_lag_2	0.153880	0.255527	0.602208	0.547036
SPF	0.565016	0.472887	1.194822	0.232156
alpha[1]	0.419351	0.260560	1.609418	0.107525
alpha[2]	0.346759	1.053861	0.329037	0.742128
beta[1]	0.058760	2.090575	0.028107	0.977577
nu	4.472871	1.614077	2.771164	0.005586
omega	0.136928	0.470167	0.291233	0.770873

GJR-GARCH:

- Model ID: `M068`
- Distribution: `$t$`
- Mean process: `(1,1)` (ARX_1_1)
- Volatility process: `GJR-GARCH(2,1)`

$$\hat{\pi}_{t+1} = 0.0846 + 0.1984\,\pi_t + 0.7847\,SPF_t + \mu_{t+1} \tag{44}$$

Robust SE: `Const` (0.0816), `Inflation_lag_1` (0.0996), `SPF` (0.1317).

$$\hat{\sigma}_t^2 = 0.1168 + 0.4252\,\left(u_{t-1}^+\right)^2 + 0.2174\,\left(u_{t-1}^-\right)^2 + 0.6575\,\left(u_{t-2}^+\right)^2 + 0.0000\,\left(u_{t-2}^-\right)^2 + 0.1354\,\sigma_{t-1}^2 \tag{45}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

- Number of observations: **215**
- Log-likelihood: **−168.657674**
- AIC: **357.315348**, BIC: **391.021728**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.084561	0.081554	1.036868	0.299797
Inflation_lag_1	0.198379	0.099598	1.991802	0.046393
SPF	0.784679	0.131672	5.959354	0.000000
alpha[1]	0.425191	0.219441	1.937612	0.052671
alpha[2]	0.657467	0.374661	1.754833	0.079288
beta[1]	0.135397	0.172961	0.782816	0.433735
gamma[1]	−0.207829	0.251448	−0.826528	0.408504
gamma[2]	−0.657467	0.305776	−2.150159	0.031543
nu	5.349660	2.200946	2.430618	0.015073
omega	0.116818	0.042038	2.778885	0.005455

EGARCH:

- Model ID: `M117`
- Distribution: `Mix of Normal`

- Mean process: `(1,1)` (ARX_1_1)
- Volatility process: `EGARCH(2,1)`

$$\hat{\pi}_{t+1} = 0.0861 + 0.2148\,\pi_t + 0.7325\,SPF_t + \mu_{t+1} \tag{46}$$

Robust SE: `Const` (0.0317), `Inflation_lag_1` (0.0705), `SPF` (0.0017).

$$\ln \hat{\sigma}_t^2 = -0.3146 + 0.5764 \left(\mid z_{t-1} \mid - \sqrt{2/\pi} \right) + 0.0053 \left(\mid z_{t-2} \mid - \sqrt{2/\pi} \right) + 0.1006\, z_{t-1} + 0.3061\, z_{t-2} + 0.6993 \ln \sigma_{t-1}^2 \tag{47}$$

- Number of observations: **215**
- Log-likelihood: **−166.237213**
- AIC: **356.474426**, BIC: **396.922083**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.086145	0.031694	2.718014	0.006568
Inflation_lag_1	0.214824	0.070517	3.046416	0.002316
SPF	0.732514	0.001739	421.149922	0.000000
alpha[1]	0.576377	0.147631	3.904164	0.000095
alpha[2]	0.005343	0.238386	0.022413	0.982118
beta[1]	0.699319	0.105518	6.627490	0.000000
gamma[1]	0.100565	0.099839	1.007278	0.313801
gamma[2]	0.306148	0.099347	3.081593	0.002059
mu_1	0.048865	0.057085	0.855992	0.392002
omega	−0.314557	0.158733	−1.981672	0.047516
p_1	0.934951	0.041199	22.693639	0.000000
sigma_1_sq	0.625460	0.155173	4.030724	0.000056

4.a.ii. *BIC:*

GARCH:

- Model ID: `M055`
- Distribution: `$t$`
- Mean process: `FC` (Constant_anchor)
- Volatility process: `GARCH(2,1)`

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{48}$$

No mean-process coefficients are estimated in this anchored specification.

$$\hat{\sigma}_t^2 = 0.1221 + 0.4616\, u_{t-1}^2 + 0.4956\, u_{t-2}^2 + 0.0000\, \sigma_{t-1}^2 \tag{49}$$

- Number of observations: **215**
- Log-likelihood: **−180.461891**
- AIC: **370.923781**, BIC: **387.776971**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.461596	0.179330	2.574004	0.010053
alpha[2]	0.495589	0.229219	2.162077	0.030612
beta[1]	0.000000	0.151356	0.000000	1.000000
nu	4.807722	1.636402	2.937984	0.003304
omega	0.122062	0.045238	2.698212	0.006971

GJR-GARCH:

- Model ID: `M050`
- Distribution: `$t$`
- Mean process: `FC` (Constant_anchor)
- Volatility process: `GJR-GARCH(1,1)`

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{50}$$

No mean-process coefficients are estimated in this anchored specification.

$$\hat{\sigma}_t^2 = 0.0636 + 0.7929\left(u_{t-1}^+\right)^2 + 0.1465\left(u_{t-1}^-\right)^2 + 0.4366\,\sigma_{t-1}^2 \tag{51}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

- Number of observations: **215**
- Log-likelihood: **−178.679065**
- AIC: **367.358130**, BIC: **384.211320**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.792917	0.256260	3.094187	0.001974
beta[1]	0.436615	0.116958	3.733109	0.000189
gamma[1]	−0.646461	0.231689	−2.790208	0.005267
nu	4.760505	1.533912	3.103507	0.001912
omega	0.063611	0.025467	2.497746	0.012499

EGARCH:

- Model ID: `M051`
- Distribution: `$t$`
- Mean process: `FC` (Constant_anchor)
- Volatility process: `EGARCH(1,1)`

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{52}$$

No mean-process coefficients are estimated in this anchored specification.

$$\ln \hat{\sigma}_t^2 = -0.2520 + 0.5595\left(\mid z_{t-1} \mid - \sqrt{2/\pi}\right) + 0.2663\, z_{t-1} + 0.7882 \ln \sigma_{t-1}^2 \tag{53}$$

- Number of observations: **215**
- Log-likelihood: **−178.479161**

- AIC: **366.958322**, BIC: **383.811512**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.559468	0.139250	4.017715	0.000059
beta[1]	0.788161	0.061181	12.882450	0.000000
gamma[1]	0.266302	0.077913	3.417950	0.000631
nu	5.151108	1.764823	2.918768	0.003514
omega	−0.251989	0.105679	−2.384477	0.017103

4.b. EGARCH and Initialization Notes

- Effective sample is fixed at 215 observations for all models (`hold_back=0` with explicit lag regressors in the mean equation).
- In `arch`, EGARCH uses the package's centered term `|z|-sqrt(2/pi)` in the recursion for all distributions.
- Under non-Gaussian errors (e.g., Student's t or mixture), this is an intercept reparameterization; fitted dynamics and likelihood are still valid.
- The recursion start (`backcast`) is updated iteratively during estimation in this script: fit -> implied initial variance from current parameters -> refit.
- Stability is monitored using a persistence proxy (<1): GARCH uses `sum(alpha)+sum(beta)`, GJR uses `sum(alpha)+0.5*sum(gamma)+sum(beta)`, EGARCH uses `sum(beta)`.

5. Regime-Switching Inflation Models

This folder estimates regime-switching models using the project effective sample and pre-created lag variables.

5.a. *Mean Process Specifications*

I estimate all four mean-process specifications from `MeanProcess/README.md`.

5.a.i. *1) Constant:*

$$\pi_t = SPF_t + u_t \tag{54}$$

with regime-switching distributional parameters.

5.a.ii. *2) ARX(1,1):*

$$\pi_t = \rho_1 \pi_{t-1} + \phi_1 SPF_t + u_t \tag{55}$$

5.a.iii. *3) ARX(2,1):*

$$\pi_t = \rho_1 \pi_{t-1} + \rho_2 \pi_{t-2} + \phi_1 SPF_t + u_t \tag{56}$$

5.a.iv. *4) ARX(2,2):*

$$\pi_t = \rho_1 \pi_{t-1} + \rho_2 \pi_{t-2} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + u_t \tag{57}$$

5.b. *Regime-Switching Structure*

Let latent state be

$$s_t \in \{1, ..., K\}, \qquad \Pr(s_t = j \mid s_{t-1} = i) = p_{ij}. \tag{58}$$

Switching settings evaluated:

Error Distribution	#Regime	Switching AR	Switching SPF	Switching Distribution
Normal / Student t	2	Y	Y	Y
Normal / Student t	2	Y	N	Y
Normal / Student t	2	N	N	Y
Normal / Student t	3	Y	N	Y
Normal / Student t	3	N	N	Y

For Student's t, `nu` is estimated and allowed to switch by regime.

Mean	Dist	K	Sw.AR	Sw.SPF	Sw.Var	Sw.nu	LogLik	AIC	BIC
ARX(1,1)	Normal	3	Y	N	Y	N	−166.916	359.833	403.651
ARX(1,1)	Student t	2	Y	Y	Y	Y	−173.779	367.559	401.265
ARX(1,1)	Student t	3	Y	N	Y	Y	−167.871	367.741	421.671
ARX(1,1)	Student t	2	Y	N	Y	Y	−174.942	367.884	398.219
ARX(1,1)	Normal	2	N	N	Y	N	−178.061	368.121	388.345
ARX(1,1)	Normal	2	Y	N	Y	N	−177.625	369.249	392.843
ARX(1,1)	Normal	3	N	N	Y	N	−174.565	371.13	408.207
ARX(1,1)	Student t	2	N	N	Y	Y	−178.362	372.723	399.689
ARX(1,1)	Student t	3	N	N	Y	Y	−176.378	380.755	427.944
ARX(1,1)	Normal	2	Y	Y	Y	N	−196.014	408.027	434.992
ARX(2,1)	Normal	3	Y	N	Y	N	−162.656	357.313	411.243
ARX(2,1)	Student t	2	Y	Y	Y	Y	−168.303	360.606	401.054
ARX(2,1)	Student t	2	Y	N	Y	Y	−172.046	366.091	403.168
ARX(2,1)	Student t	3	Y	N	Y	Y	−164.587	367.175	431.217
ARX(2,1)	Normal	2	N	N	Y	N	−176.866	367.731	391.325
ARX(2,1)	Normal	3	N	N	Y	N	−172.054	368.108	408.555
ARX(2,1)	Student t	2	N	N	Y	Y	−176.331	370.663	400.998
ARX(2,1)	Normal	2	Y	N	Y	N	−176.535	371.07	401.406
ARX(2,1)	Normal	2	Y	Y	Y	N	−176.471	372.942	406.648
ARX(2,1)	Student t	3	N	N	Y	Y	−174.162	378.325	428.884
ARX(2,2)	Student t	3	Y	N	Y	Y	−162.901	365.803	433.216
ARX(2,2)	Student t	2	Y	N	Y	Y	−171.661	367.322	407.769
ARX(2,2)	Normal	3	Y	N	Y	N	−166.939	367.877	425.178
ARX(2,2)	Normal	2	N	N	Y	N	−176.563	369.126	396.091
ARX(2,2)	Normal	3	N	N	Y	N	−171.804	369.608	413.426
ARX(2,2)	Student t	2	Y	Y	Y	Y	−171.537	371.073	418.262
ARX(2,2)	Student t	2	N	N	Y	Y	−175.668	371.335	405.042
ARX(2,2)	Normal	2	Y	N	Y	N	−176.174	372.348	406.055
ARX(2,2)	Normal	2	Y	Y	Y	N	−176.143	376.286	416.733
ARX(2,2)	Student t	3	N	N	Y	Y	−173.456	378.912	432.843
Constant	Normal	3	N	N	Y	N	−180.655	379.311	409.646
Constant	Normal	2	N	N	Y	N	−186.63	381.26	394.743
Constant	Student t	2	N	N	Y	Y	−188.839	389.678	409.902
Constant	Student t	3	N	N	Y	Y	−184.478	392.955	433.403

6. BEGE Density

BEGE density function $f_{BEGE}(\mu \mid p, n, \sigma_p, \sigma_n)$ is the function that calculates the density of the observation μ given the parameters $\{p, n, \sigma_p, \sigma_n\}$.

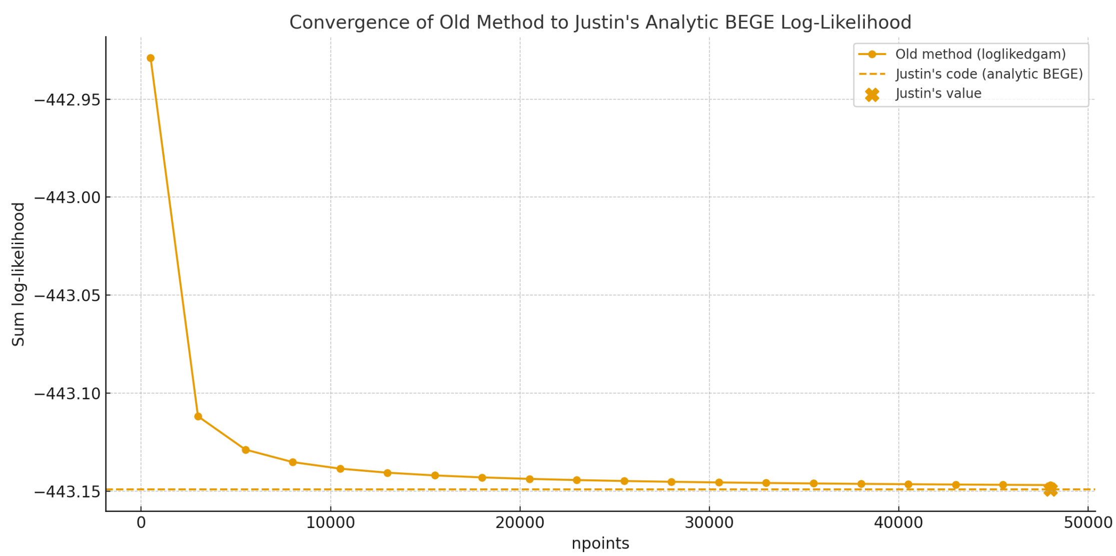
To compare Justin's analytic BEGE density code with the old numerical one, I conducted an experiment using synthetic data. I generated 200 independent observations from a standard normal distribution and fixed the BEGE parameters to

$$p = 5, \qquad n = 7, \qquad \sigma_p = 0.8, \qquad \sigma_n = 1.2. \tag{59}$$

I computed the sum of log-likelihood using two approaches:

1. The *old BEGE function*, which approximates the density via numerical integration on a uniform grid, and whose accuracy depends strongly on the number of grid points; and
2. *Justin's analytic BEGE function.*

The old numerical method was evaluated over a wide range of grid resolutions (up to $50,000$ points), and the resulting log-likelihood sums were plotted alongside the benchmark value computed from Justin's code.



Findings:

- The old numerical likelihood converges toward Justin's analytic likelihood as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood. This explains why, for a given set of parameters, Justin's implementation produces a lower log-likelihood than the old code.
- Justin's analytic BEGE formula provides a more accurate and more computationally efficient evaluation.

7. BEGE GARCH Family

To start the random search of initial mean parameters, I draw uniform samples from $(\mu - 2\sigma,\ \mu + 2\sigma)$ where μ and σ are mean and standard deviation from OLS regression. I also set the AR coefficient bound to avoid the AR process to explode. We have four types of mean processes.

Table 1: Parameter Bounds for Mean Process Specifications

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	–	–	–	–	–
ARX(1,1)	$(\min \pi_t,\ \ \max \pi_t)$	$(-0.999,\ 0.999)$	–	$(-10,\ 10)$	–
ARX(2,1)	$(\min \pi_t,\ \ \max \pi_t)$	$(-1.999,\ 1.999)$	$(-0.999,\ 0.999)$	$(-10,\ 10)$	–
ARX(2,2)	$(\min \pi_t,\ \ \max \pi_t)$	$(-1.999,\ 1.999)$	$(-0.999,\ 0.999)$	$(-10,\ 10)$	$(-10,\ 10)$

7.a. Constant BEGE

The first model that I estimated is BEGE with invariant shape parameters, which is

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1} \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n \end{aligned} \tag{60}$$

where

$$\omega_p \sim \tilde{\Gamma}(\bar{p}, 1), \quad \omega_n \sim \tilde{\Gamma}(\bar{n}, 1). \tag{61}$$

$\bar{p}$ and $\bar{n}$ are unconditional shape parameters. The random search and bounds are set to be:

- $0.05 < \sigma_p, \sigma_n < 2$
- $0.1 < p, n < 10$

7.b. Symmetric BEGE

The model assumes different scaled parameters and different constants in GJR recursions for both components:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \, \omega_p - \sigma_n \, \omega_n, \end{aligned} \tag{62}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \tag{63}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho, \phi^+, \phi^-)$:

$$\begin{aligned} p_t &= p_0 + \rho \, p_{t-1} + \frac{\phi^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_p^2} \left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho \, n_{t-1} + \frac{\phi^+}{2\sigma_n^2} \left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2 \end{aligned} \tag{64}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho < 0.999$,
- $10^{-5} < \phi^+, \phi^- < 0.999$,

- $10^{-5} < \sigma_p, \sigma_n < 2$.

7.c. Bad and Good Symmetric GARCH Model

Here n_t, p_t is generated recursively with parameters $\left(p_0, n_0, \rho_p, \rho_n, \phi_p, \phi_n\right)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p}{2\sigma_p^2}\left(u_{t-1}\right)^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n}{2\sigma_n^2}\left(u_{t-1}\right)^2 \end{aligned} \tag{65}$$

I have set the constraints below.

- $\rho + \phi < 1$ for both good and bad shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$.
- $\max\{p_t, n_t\} < 200$

7.d. Inflation and Deflation Model

Here n_t, p_t is generated recursively with parameters $\left(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_n^-\right)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2 \end{aligned} \tag{66}$$

According to the emails, I have set the constraints below.

- $\rho_p + \frac{\phi_p^+}{2} < 1$ and $\rho_n + \frac{\phi_n^-}{2} < 1$.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$.
- $\max\{p_t, n_t\} < 200$

7.e. Full BEGE

BEGE GARCH has mean process and higher:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p\, \omega_p - \sigma_n\, \omega_n, \end{aligned} \tag{67}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \tag{68}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $\left(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^-\right)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2 + \frac{\phi_p^-}{2\sigma_p^2}\left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2}\left(u_{t-1}^+\right)^2 + \frac{\phi_n^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2 \end{aligned} \tag{69}$$

The parameter bounds are chosen to be:

- $0.005 < p_0, n_0 < 10$,
- $10^{-5} < \rho_p, \rho_n < 0.999$,

- $10^{-5} < \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^- < 0.999$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

According to the emails, I have set the constraints below.

- $\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1$ for both BE and GE shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$ where $\mathrm{Var}(\pi_t) = 0.75$ is the unconditional variance of inflation.
- $\max\{p_t, n_t\} < 200$.

8. Result Summary

Overall, most of the estimations with random initial values converge. I randomly sampled $100,000$ starting values for the constant BEGE model, and sampled $10,000$ for symmetric BEGE because it is more time consuming. I report the best estimation below.